CHANDU PRASAD BHAIRAPU



AI & Process Automation • AI Assistants & Agents

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ABOUT ME

AI engineer (MSc in Artificial Intelligence, 3 years of experience) who builds AI assistants, agents and automation workflows, writes their prompts and decision logic, and tests them against ground truth until people can rely on them. Three roles at Covestro in the chemical industry, two of them working directly with chemists and their data; most recently sole developer of an agentic knowledge-extraction system for the German energy sector, built with SAP domain experts, raising precision/recall from 33%/45% to 93%/90%.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer of an AI-based knowledge-extraction workflow for the German energy sector, built end-to-end (data ingestion, chunking, prompt engineering, indexing with agentic, deterministic, and monolithic strategies); reported directly to the CTO and worked with SAP domain experts across the German energy sector.
- Built and tested AI agents with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) against an SAP-system-derived ground truth, raising precision/recall from 33%/45% to 93%/90%.
- Engineered a deterministic (code-based) pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Developed Expert Interface, an internal web tool (React, Python microservices) on Azure that lets domain experts review and approve changes to the knowledge base themselves.
- Built a Model Context Protocol (MCP) service that gives AI assistants millisecond access to the knowledge graph, integrated into Claude Code for agentic development workflows.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, prompt engineering, MCP, Claude Code, RAG, GraphRAG, FastAPI, React, Neo4j, MongoDB, Postgres, Docker, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Built an AI agent workflow with LangGraph whose decision logic routes each request to vector database search, general inference, or summarization, and deployed it on AWS.
- Built the user interface (React) and the Python/FastAPI backend; containerized with Docker, deployed with AWS CDK, versioned in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, React, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated the testing of an AI assistant over Covestro's internal chemical data (RAG) with RAGAS and PytrecEval, comparing its answers to ground truth written by chemists.
- Benchmarked multilingual embedding models on over 1M chunks of internal chemical documents to find the best retrieval setup.
- Built the evaluation infrastructure on AWS (Lambda, SQS, DynamoDB) with CDK.
- **Tech stack: Python, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, S3, CDK), Azure AI Search**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Analysed time-series process data and built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity), improving early identification of batches likely to deviate from specification.
- Built a Streamlit dashboard letting chemists simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Pandas, Scikit-Learn, XGBoost, TensorFlow/Keras, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.

PROJECTS

Project: Product Q&A Assistant | n8n | RAG | Python | LLM APIs

- n8n automation workflow that sorts incoming customer and retailer emails, answers product and application questions from technical and safety data sheets with a RAG-based AI assistant, and routes uncertain cases to a person for approval.
- Prompts and decision logic are versioned and tested on a labelled question set; personal data is removed before any LLM call (GDPR), answers are marked as AI-generated, and the process documentation includes notes on EU AI Act transparency and GDPR.

TECHNICAL SKILLSET

- **AI & Automation:** automation workflows in Python, AI assistants and agents (LangGraph, LangChain), prompt engineering, decision logic, RAG, LLM APIs (LiteLLM), AI output evaluation (RAGAS, PytrecEval), MCP, Claude Code
- **Data Analysis:** Python, Pandas, NumPy, SQL, Scikit-Learn, XGBoost, time-series analysis, Matplotlib, Streamlit
- **Software & Cloud:** FastAPI, REST APIs, React (beginner), Docker, GitLab CI/CD, AWS, Azure
- **Databases:** Postgres, MongoDB, Neo4j, vector databases (Qdrant, Pinecone, Azure AI Search)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering, G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT
- Udemy - The Complete Neural Networks Boot camp

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)